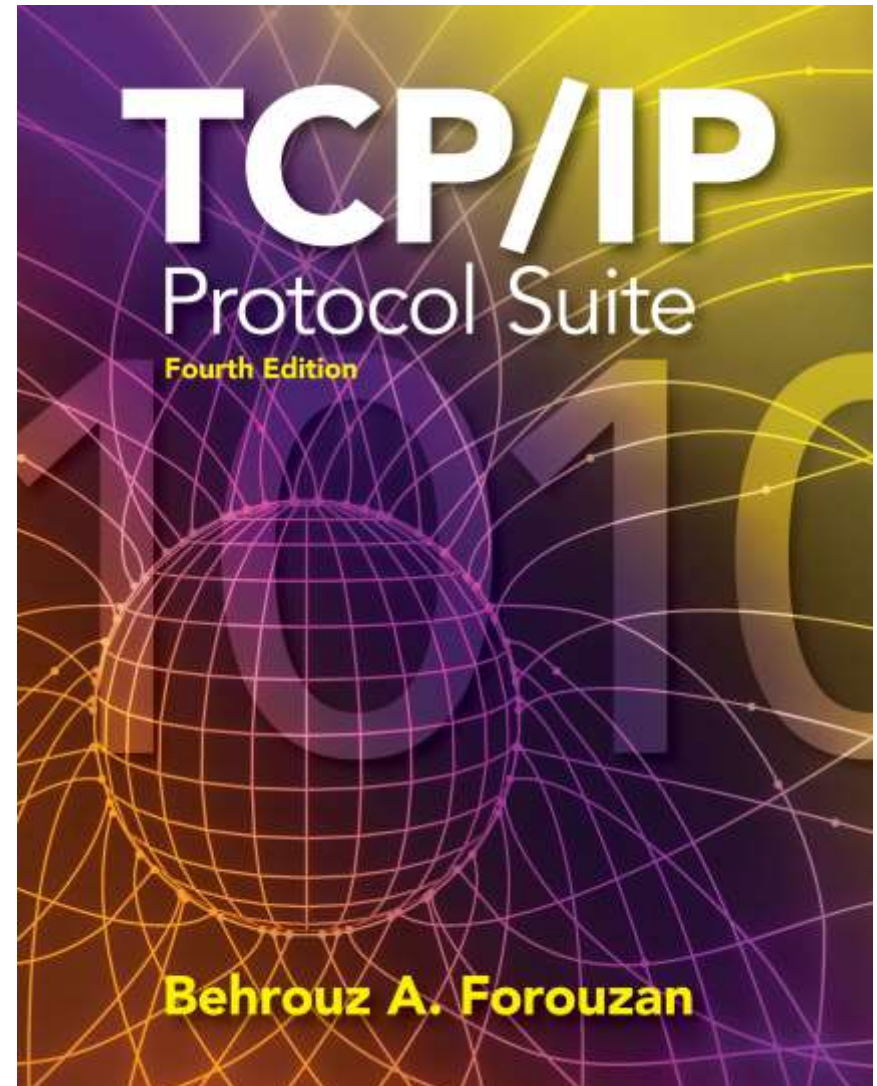


Chapter 1

Introduction



OBJECTIVES:

- ☐ To give a brief history of the Internet.
- ☐ To give the definition of the two often-used terms in the discussion of the Internet: protocol and standard.
- ☐ To categorize standard organizations involved in the Internet and give a brief discussion of each.
- ☐ To define Internet Standards and explain the mechanism through which these standards are developed.
- ☐ To discuss the Internet administration and give a brief description of each branch.

Chapter Outline

1.1 A Brief History

1.2 Protocols and Standards

1.3 Standards Organizations

1.4 Internet Standards

1.5 Internet Administration

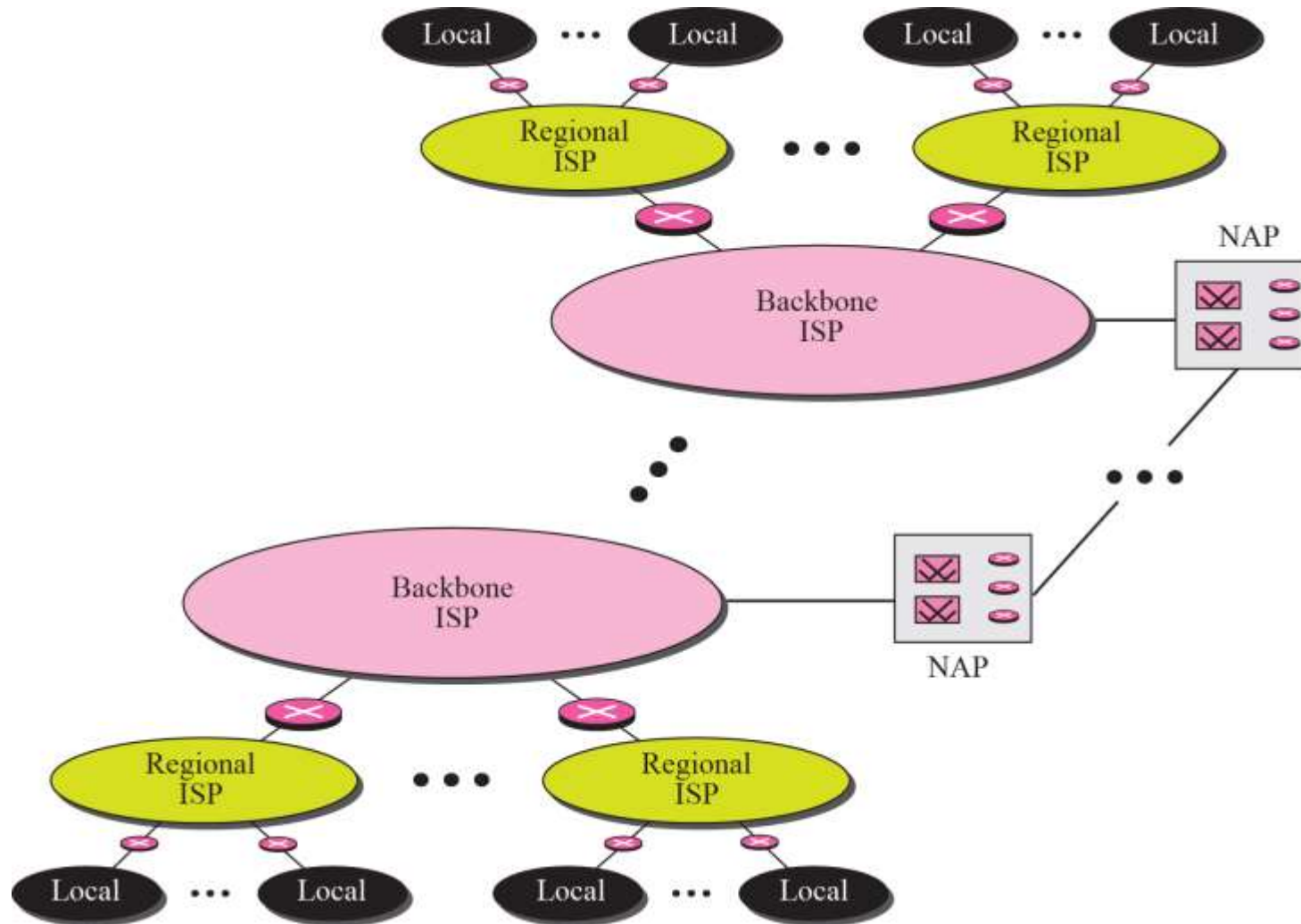
1-1 A BRIEF HISTORY

- A network is a group of connected, communicating devices such as computers and printers.
- An *internet* is two or more networks that can communicate with each other.
- The most notable internet is called *the internet*, composed of hundreds of thousands of interconnected networks.
- Private individuals as well as various organizations such as government agencies, schools, research facilities, corporations, and libraries in more than 100 countries use the Internet.

Topics Discussed in the Section

- ✓ **ARPANET**
- ✓ **Birth of the Internet**
- ✓ **TCP/IP**
- ✓ **MILNET**
- ✓ **CSNET**
- ✓ **NSFNET**
- ✓ **ANSNET**
- ✓ **The Internet Today**
- ✓ **World Wide Web**
- ✓ **Growth of the Internet**

Figure 1.1 *Internet today*



1-2 PROTOCOLS AND STANDARDS

- In this section, we define two widely used terms: protocols and standards.
- First, we define *protocol*, which is synonymous with “rule.”
- Then we discuss *standards*, which are agreed-upon rules.

Topics Discussed in the Section

- ✓ **Protocols**
- ✓ **Standards**

1-3 STANDARDS ORGANIZATION

Standards are developed through the cooperation of standards creation committees, forums, and government regulatory agencies.

Topics Discussed in the Section

- ✓ **Standards Creation Committees**
- ✓ **Forums**
- ✓ **Regulatory Agencies**

1-4 INTERNET STANDARDS

- **An Internet standard is a thoroughly tested specification that is useful to and adhered to by those who work with the Internet.**
- **It is a formalized regulation that must be followed. There is a strict procedure by which a specification attains Internet standard status.**
- **A specification begins as an Internet draft. An Internet draft is a working document with no official status and a six-month lifetime.**

Topics Discussed in the Section

- ✓ **Maturity Levels**
- ✓ **Requirement Levels**

Figure 1.2 *Maturity levels of an RFC*

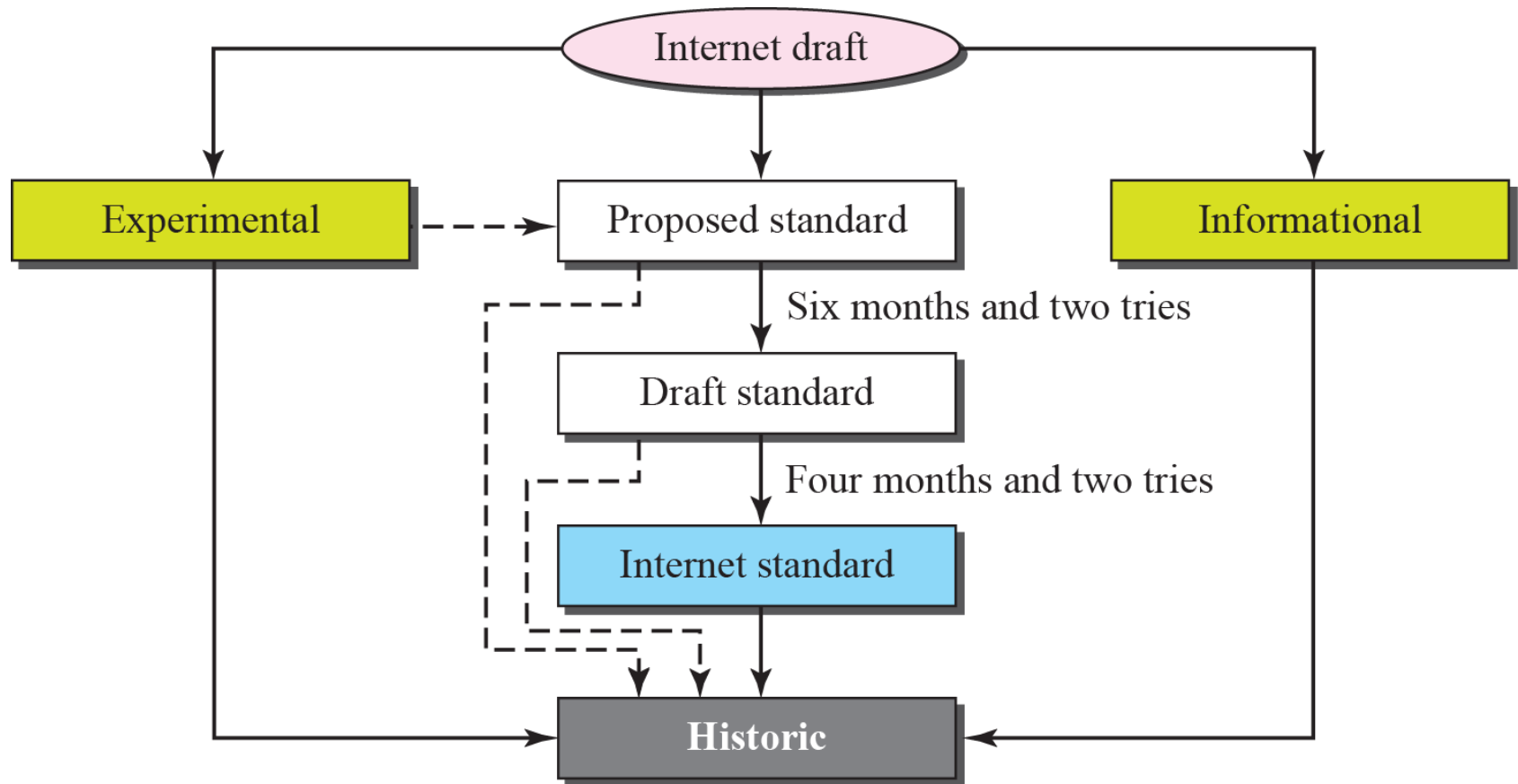
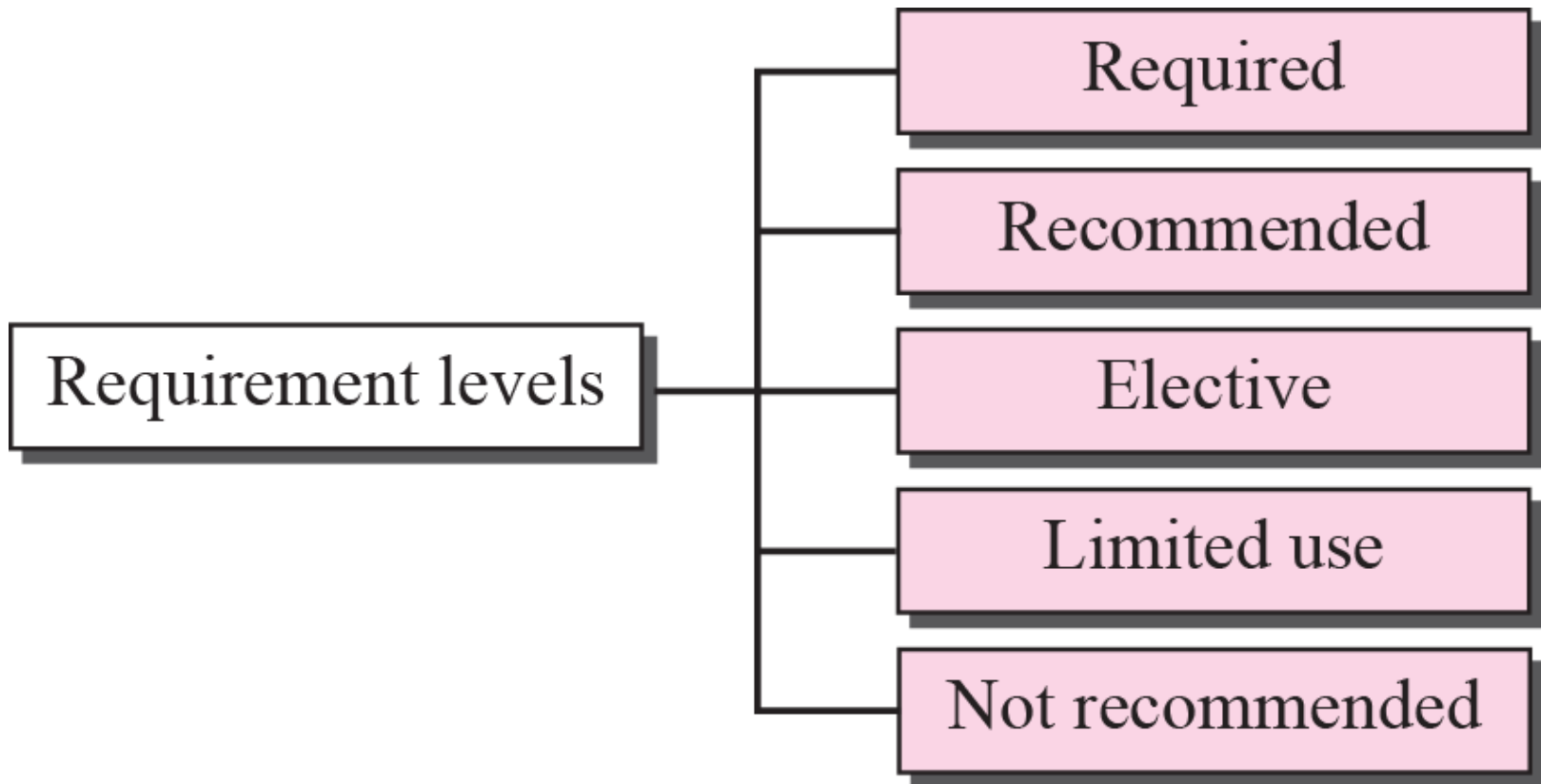


Figure 1.3 *Requirement levels of an RFC*





Note

*RFCs can be found at
<http://www.rfc-editor.org>.*

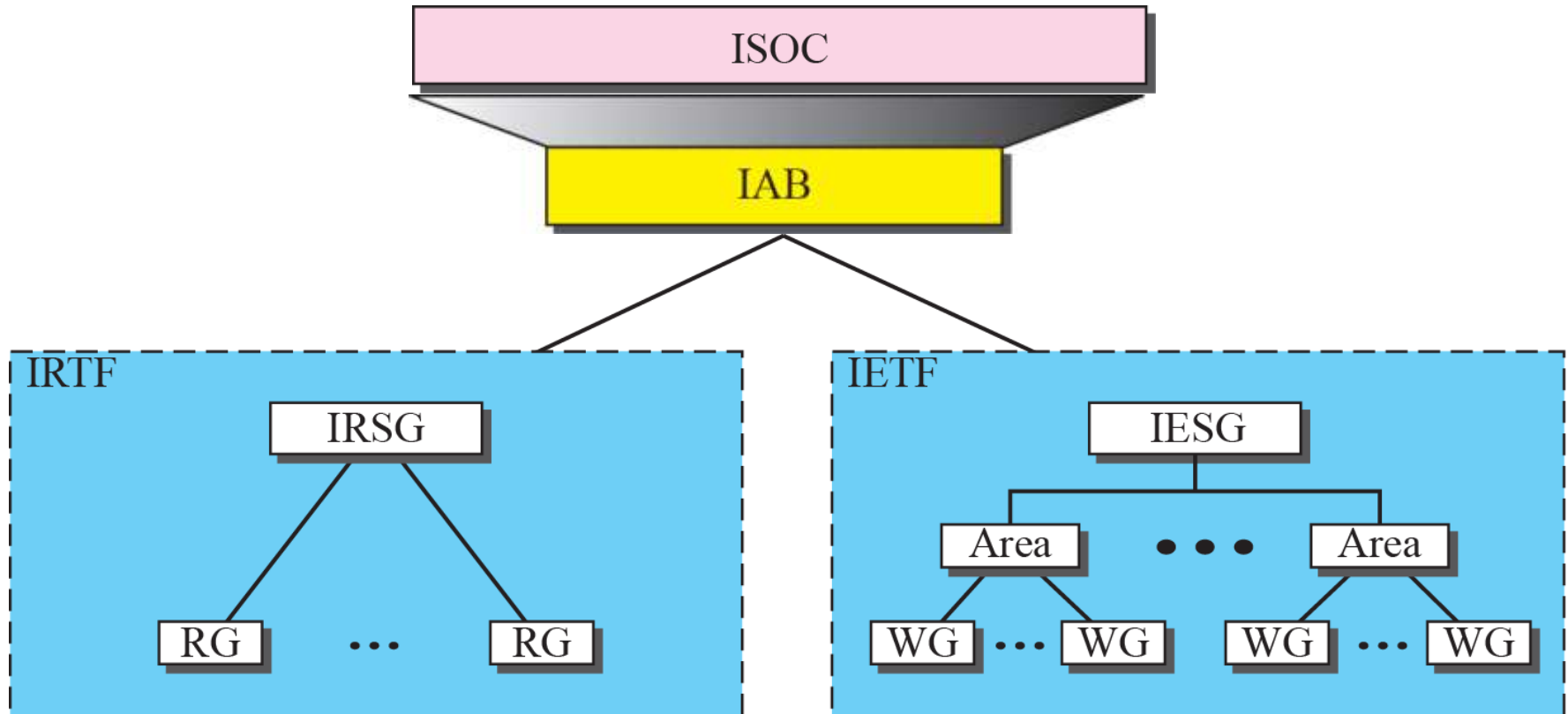
1-5 INTERNET ADMINISTRATION

- **The Internet, with its roots primarily in the research domain, has evolved and gained a broader user base with significant commercial activity.**
- **Various groups that coordinate Internet issues have guided this growth and development.**
- **Appendix G gives the addresses, e-mail addresses, and telephone numbers for some of these groups.**
- **Figure 1.4 shows the general organization of Internet administration.**

Topics Discussed in the Section

- ✓ **Internet Society (ISOC)**
- ✓ **Internet Architecture Board (IAB)**
- ✓ **Internet Research Task Force (IRTF)**
- ✓ **Internet Assigned Number Authority (IANA)**
- ✓ **Internet Corporation for Names and Numbers (ICANN)**
- ✓ **Network Information Center (NIC)**

Figure 1.4 *Internet administration*



Note

The addresses and websites for Internet organizations can be found in Appendix G.